DATA ANALYTICS FOR BUSINESS

with Lizhen Xu, Ph.D.

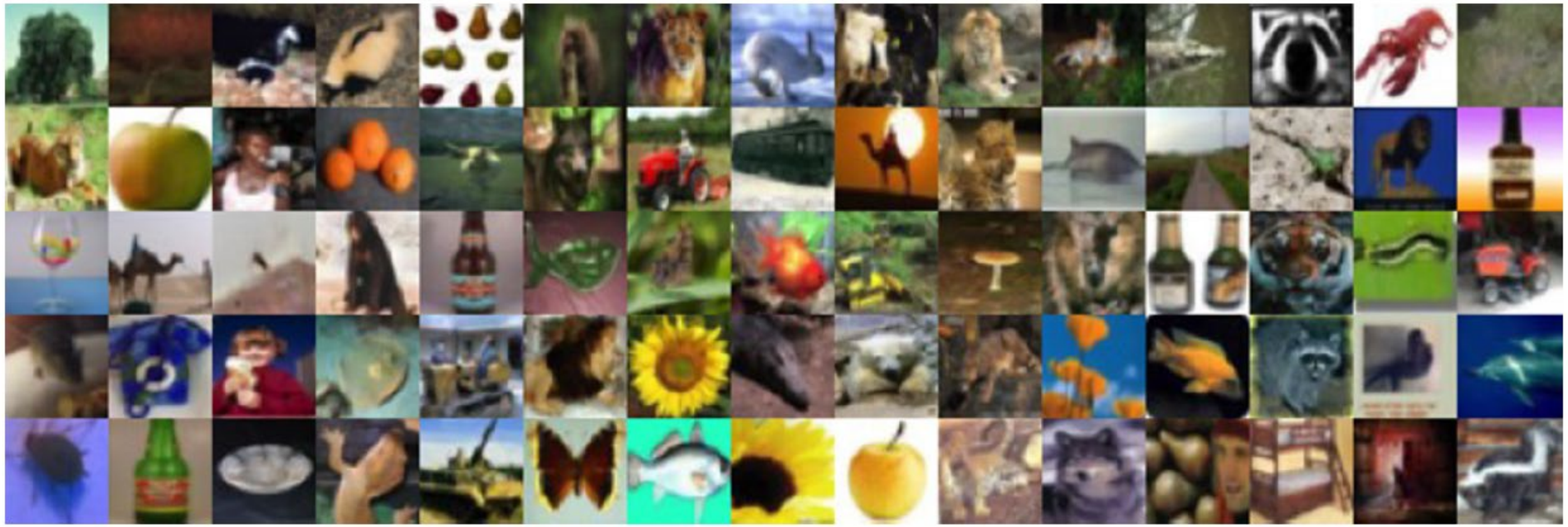
Deep Learning

Learning Objectives

- Describe the architecture of Convolutional Neural Networks (CNN)
  - Explain Convolution
  - Explain Pooling
- Perform CNN for image classification using deep learning software tools

Convolutional Neural Networks (CNN)

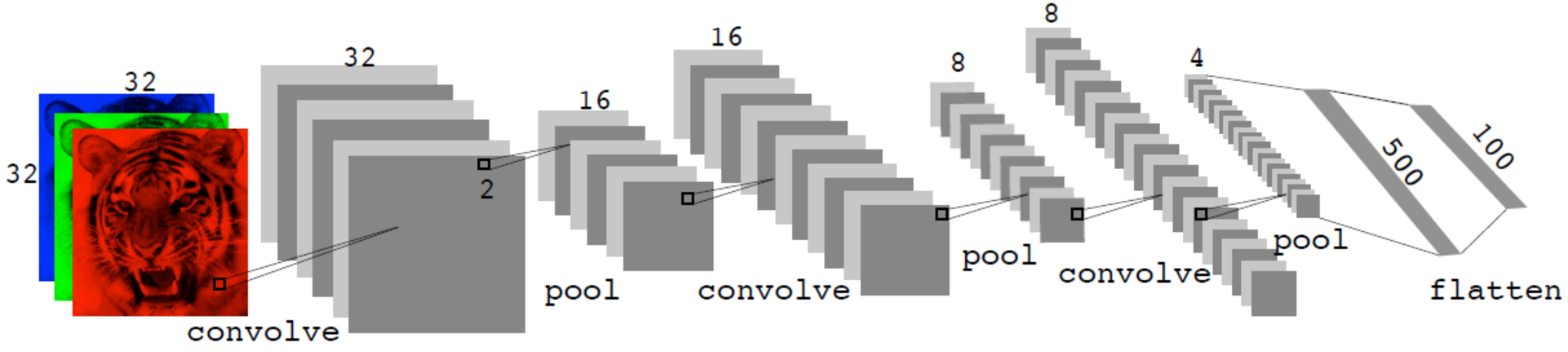
- Image Classification
  - Color pictures: three 8-bit numbers per pixel, for red, green, and blue
  - A large number of classes
  - CIFAR-100 database*



*Reference: Krizhevsky A (2009) Learning multiple layers of features from tiny images. University of Toronto, Tech. (https://www.cs.toronto.edu/~kriz/learning-features-2009-TR.pdf)

Convolutional Neural Networks (CNN) (Cont'd)

- Convolutional Neural Networks (CNN)
  - First identify low-level features
  - Combine low-level features to form higher-level features
  - Based on the presence/absence of higher-level features to predict the probability of class membership



James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An introduction to statistical learning: with applications in R*. Springer. 2nd Ed. p 416.

Convolution

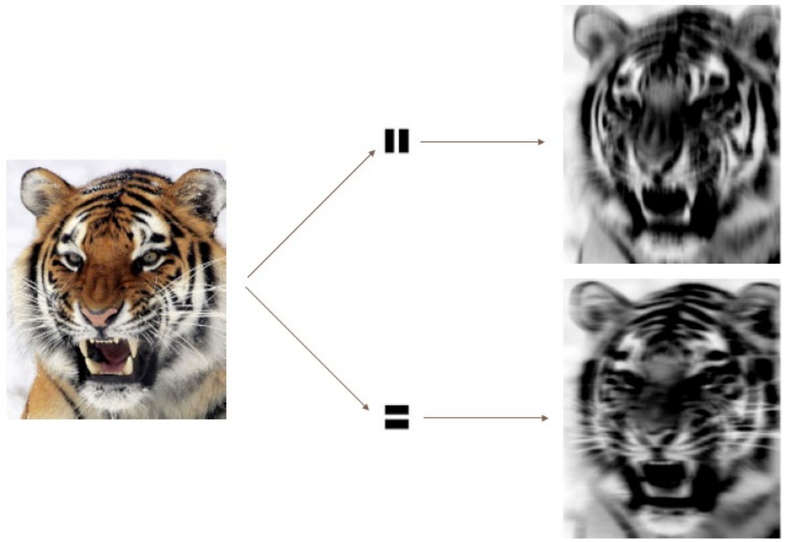
- Convolution
  - Determine whether a particular local feature is present in an image
  - Repeatedly multiply matrix elements and then add them together
  - Usually use a large number of *convolution filters*

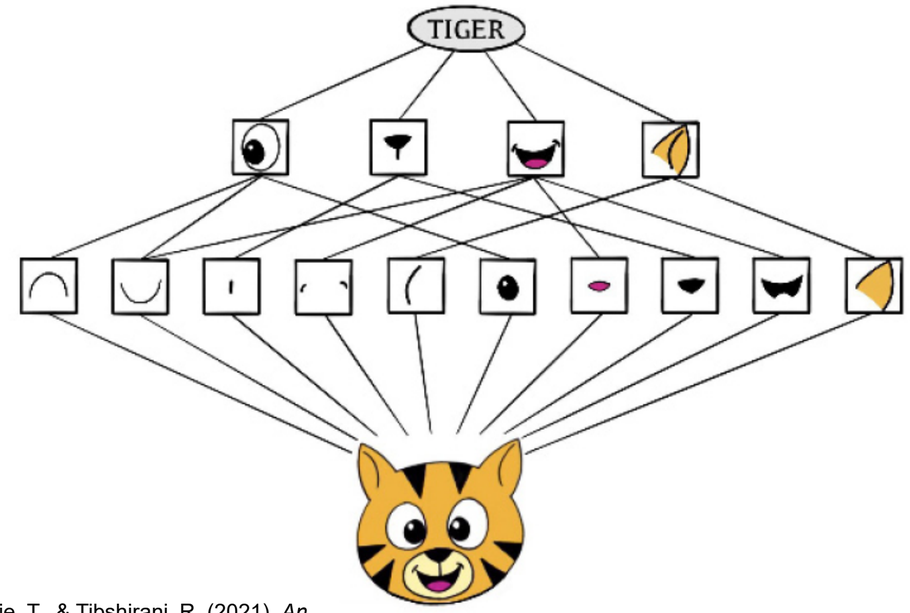
$$\text{Original Image} = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \\ j & k & l \end{bmatrix} \qquad \text{Convolution Filter} = \begin{bmatrix} \alpha & \beta \\ \gamma & \delta \end{bmatrix}$$

$$\text{Convolved Image} = \begin{bmatrix} a\alpha + b\beta + d\gamma + e\delta & b\alpha + c\beta + e\gamma + f\delta \\ d\alpha + e\beta + g\gamma + h\delta & e\alpha + f\beta + h\gamma + i\delta \\ g\alpha + h\beta + j\gamma + k\delta & h\alpha + i\beta + k\gamma + l\delta \end{bmatrix}$$

Convolution (Cont'd)

- Convolution
  - Regions of the original image that *resemble* the convolution filter are *highlighted* in the convolved image;
  - Regions of the original image that *differ* from the convolution filter are *weakened* in the convolved image.
  - A bank of *predefined* filters in image processing
  - Filter weights are usually *learned* in CNN





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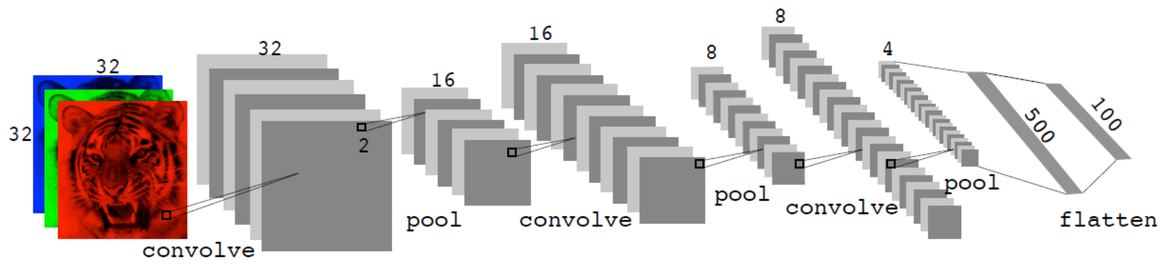
Pooling

- Pooling
  - Condense a large image into a smaller summary image
  - *Max pooling*: summarize each non-overlapping 2x2 block of pixels using the maximum value in the block
  - Reduce the size of an image by a factor of four
  - Pooling reduces dimensionality while providing *location invariance*

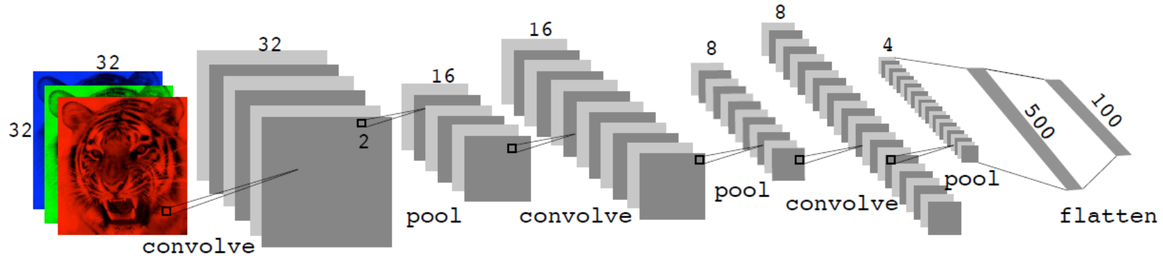
$$\text{Max pool} \begin{bmatrix} 1 & 2 & 5 & 3 \\ 3 & 0 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 1 & 2 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 3 & 5 \\ 2 & 4 \end{bmatrix}$$

CNN Architecture



- Input: a three-dimensional *feature map* (three *channels*: one for each color)
- Convolution filter:
  - Each filter has the *same number of channels* as the input feature map
  - Results from all channels of a filter are *summed* to form a 2D feature map
- Convolution output: a three-dimensional feature map with the *number of channels equal to the number of filters*
  - *Padding* at the edges
- Apply ReLU activation function to the convolved image, also called *detector layer*.
- Apply Max Pooling to the convolution output, which completes one convolution-pooling cycle.

CNN Architecture (Cont'd)



- Repeat the convolution-pooling cycles.
- *Increase the number of filters* in the next convolve layer to compensate the smaller dimension of the pooled images.
- *Flatten*: treat all pixels as separate units
- One or more *fully-connected layers* before reaching the output layer
- *Softmax* activation function in the output layer for the final classification output.
- May apply *Ridge or Lasso Regularization* and *Dropout Learning* to reduce the number of parameters.

Software Tools for Deep Learning

- Tensorflow
  - Deep learning platform developed by Google in *Python* language
  - *Keras*: high-level interface with *Tensorflow*
  - *Keras* and *Tensorflow* available in R package
  - Communicate via "*reticulate*"
  - Require the setup of Python to use the *Keras* package in R

- PyTorch
  - With both *Python* and *C++* interfaces
  - *Torch* package in R links directly to the C++ backend
  - *Luz* package in R provides high-level APIs for *Torch*
  - Easy to set up in R, just as regular R packages

Summary

- Architecture of Convolutional Neural Networks (CNN)
  - Convolution
  - Pooling
- Apply CNN for image classification
  - PyTorch